

BUSINESS REQUIREMENT DOCUMENT (BRD)

Enterprise Manufacturing Data Platform (EMDP)

1. Background

ABC Manufacturing Ltd. operates five smart factories where production, quality, ERP, warehouse, inventory, maintenance, and IoT systems generate large volumes of data. The data is currently stored in isolated systems and formats, limiting operational visibility and making organization-wide decision-making more difficult.

2. Business Problem Statement

Operational data generated across the five factories is fragmented across heterogeneous systems and formats. There is no unified and trusted source of data for analytics and reporting. As a result, plant and executive stakeholders face difficulty obtaining a timely and accurate view of production performance, quality, inventory health, and related operational information.

3. Business Objectives

- Collect manufacturing data from multiple heterogeneous sources.
- Cleanse and validate incoming data before it is used for reporting.
- Store curated and trusted datasets in an enterprise PostgreSQL data warehouse.
- Maintain metadata and end-to-end data lineage for governance and auditability.
- Support analytics and reporting through business-ready dashboards.
- Use Git for collaborative, version-controlled development.
- Produce enterprise-grade documentation throughout the project lifecycle.

4. Stakeholders

Role	Responsibility
Data Engineering Lead	Project guidance, architecture sign-off, and sprint reviews.
Graduate Data Engineer / Project Team	Design, build, test, document, and deliver the data platform.
Plant / Production Managers	Provide production and quality data requirements and validate operational needs.
ERP / IT Administrator	Provide access to ERP, inventory, maintenance, and related enterprise systems.
Business Analysts	Define reporting KPIs and dashboard requirements.
Quality Assurance Team	Define data quality rules and acceptance

	criteria.
Executive Sponsors	Use production intelligence dashboards to support business decision-making.

5. Data Sources Identified

Source System	Format	Data Type
Production System (MES)	CSV / SQL	Machine output, downtime, cycle times
Quality Management System	Excel / CSV	Inspection results, defect records
ERP System	SQL / XML	Purchase orders, materials, finance
Warehouse Management System	CSV	Stock movement, dispatch records
Inventory System	Excel	Stock levels, reorder points
Maintenance System (CMMS)	JSON / CSV	Work orders, equipment logs
IoT Sensor Gateway	JSON	Machine telemetry, sensor readings

6. Project Scope

6.1 In Scope

- Batch ingestion, staging, cleansing, and warehousing of manufacturing data from all five factories.
- Data quality validation, metadata management, and lineage documentation.
- Star-schema data warehouse design for production intelligence reporting.
- Power BI executive dashboards.
- Git-based version control and enterprise project documentation.

6.2 Out of Scope

- Real-time or streaming data ingestion; the project focuses on batch processing.
- Predictive or prescriptive machine learning models.
- Mobile application development.

7. Functional Requirements

- The system shall ingest data from CSV, Excel, JSON, XML, and SQL sources.
- The system shall stage raw data without transformation in the Bronze layer.
- The system shall profile and cleanse data before warehousing in the Silver layer.
- The system shall organize curated data into a dimensional Star Schema in the Gold layer.
- The system shall log ETL executions and data quality exceptions.
- The system shall maintain a Data Dictionary, Business Glossary, and data lineage documentation.

- The system shall expose curated data to Power BI for reporting.

8. Non-Functional Requirements

- All source code and configuration shall be version-controlled using Git/GitHub.
- ETL pipelines shall be re-runnable and idempotent for a given batch.
- Data loads shall be auditable using batch ID, timestamp, and source file information.
- Project documentation shall be maintained and updated at the end of every sprint.

9. Technology Requirements

Category	Technology
ETL	Pentaho Data Integration (Spoon)
Database	PostgreSQL
Programming / Profiling	Python (Pandas)
Version Control	Git & GitHub
Reporting	Power BI
Documentation	Markdown / MS Word
Project Management	Agile Scrum

10. Assumptions and Constraints

- Representative sample data will be available for each identified source system.
- PostgreSQL, Pentaho, Python, Git, and Power BI are available in the project environment.
- The project follows a three-month Agile Scrum delivery model across four sprints.

11. Success Criteria

- Data from the identified source systems is successfully ingested and staged.
- Data quality checks are performed and documented for each dataset.
- A working PostgreSQL data warehouse with a Star Schema is delivered.
- Metadata and data lineage are documented and traceable.
- A business-ready Power BI dashboard is delivered and demonstrated.
- Project code, configuration, and documentation are maintained in Git/GitHub.

12. Sprint 0 Acceptance

Sprint 0 is considered complete when the business problem, stakeholders, requirements, scope, high-level architecture, project repository, folder structure, backlog, and project charter have been documented and reviewed.